

COVID-19 Global Trends Analysis

Objective

The objective of this project is to analyze COVID-19 trends over time using publicly available data. The analysis focuses on cases, deaths, and comparisons between countries to identify patterns and trends.

Dataset Description

The dataset used in this project contains global COVID-19 statistics, including daily and cumulative cases and deaths. The data covers multiple countries and time periods, allowing trend analysis and country-level comparisons.

```
import pandas as pd

url = "https://raw.githubusercontent.com/owid/covid-19-data/master/public/data/jhu/full_data.csv"

df = pd.read_csv(url)

print(df.head())
```

	date	location	new_cases	new_deaths	total_cases	total_deaths
0	2020-02-24	Afghanistan	5.0	NaN	5.0	NaN
1	2020-02-25	Afghanistan	0.0	NaN	5.0	NaN
2	2020-02-26	Afghanistan	0.0	NaN	5.0	NaN
3	2020-02-27	Afghanistan	0.0	NaN	5.0	NaN
4	2020-02-28	Afghanistan	0.0	NaN	5.0	NaN

	weekly_cases	weekly_deaths	biweekly_cases	biweekly_deaths
0	NaN	NaN	NaN	NaN
1	NaN	NaN	NaN	NaN
2	NaN	NaN	NaN	NaN
3	NaN	NaN	NaN	NaN
4	NaN	NaN	NaN	NaN

```
print("Rows and Columns:")
print(df.shape)
```

```
Rows and Columns:
(248346, 10)
```

```
print(df.columns)
```

```
Index(['date', 'location', 'new_cases', 'new_deaths', 'total_cases',
      'total_deaths', 'weekly_cases', 'weekly_deaths', 'biweekly_cases',
      'biweekly_deaths'],
      dtype='object')
```

```
df.isnull().sum()
```

	0
date	0
location	0
new_cases	271
new_deaths	19838
total_cases	9
total_deaths	19754
weekly_cases	1475
weekly_deaths	21024
biweekly_cases	3228
biweekly_deaths	22793

dtype: int64

Missing Value Analysis

The dataset contains some missing values (NaN). These missing values occur because certain countries did not report data for every metric on every date. Missing values are common in real-world datasets and were handled appropriately during the analysis.

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 248346 entries, 0 to 248345
Data columns (total 10 columns):
#   Column              Non-Null Count  Dtype
---  -
0   date                248346 non-null object
1   location            248346 non-null object
2   new_cases           248075 non-null float64
3   new_deaths          228508 non-null float64
4   total_cases         248337 non-null float64
5   total_deaths        228592 non-null float64
6   weekly_cases        246871 non-null float64
7   weekly_deaths       227322 non-null float64
8   biweekly_cases      245118 non-null float64
9   biweekly_deaths     225553 non-null float64
dtypes: float64(8), object(2)
memory usage: 18.9+ MB
```

```
df['date'] = pd.to_datetime(df['date'])
```

```
print(df.dtypes)
```

```
date                datetime64[ns]
location            object
new_cases           float64
new_deaths          float64
total_cases         float64
total_deaths        float64
weekly_cases        float64
weekly_deaths       float64
biweekly_cases      float64
biweekly_deaths     float64
dtype: object
```

```
countries = ['Pakistan', 'India', 'United States']
```

```
selected = df[df['location'].isin(countries)]
```

```
selected.head()
```

	date	location	new_cases	new_deaths	total_cases	total_deaths	weekly_cases	weekly_deaths	biweekly_cases	biweekly_deaths
104095	2020-01-30	India	1.0	NaN	1.0	NaN	NaN	NaN	NaN	NaN
104096	2020-01-31	India	0.0	NaN	1.0	NaN	NaN	NaN	NaN	NaN
104097	2020-02-01	India	0.0	NaN	1.0	NaN	NaN	NaN	NaN	NaN

Next steps:

[Generate code with selected](#)
[New interactive sheet](#)

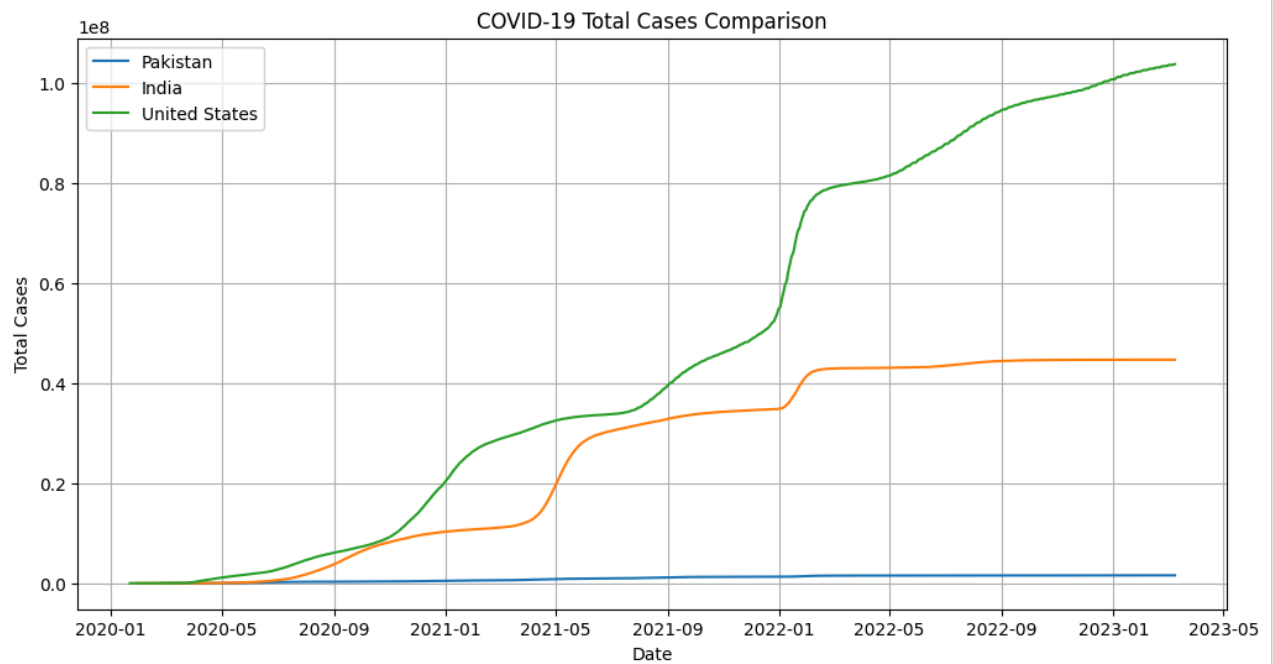
```
import matplotlib.pyplot as plt

plt.figure(figsize=(12,6))

for country in countries:
    data = selected[selected['location'] == country]
    plt.plot(data['date'], data['total_cases'], label=country)

plt.title('COVID-19 Total Cases Comparison')
plt.xlabel('Date')
plt.ylabel('Total Cases')
plt.legend()
plt.grid(True)

plt.show()
```



✓ Total Cases Comparison

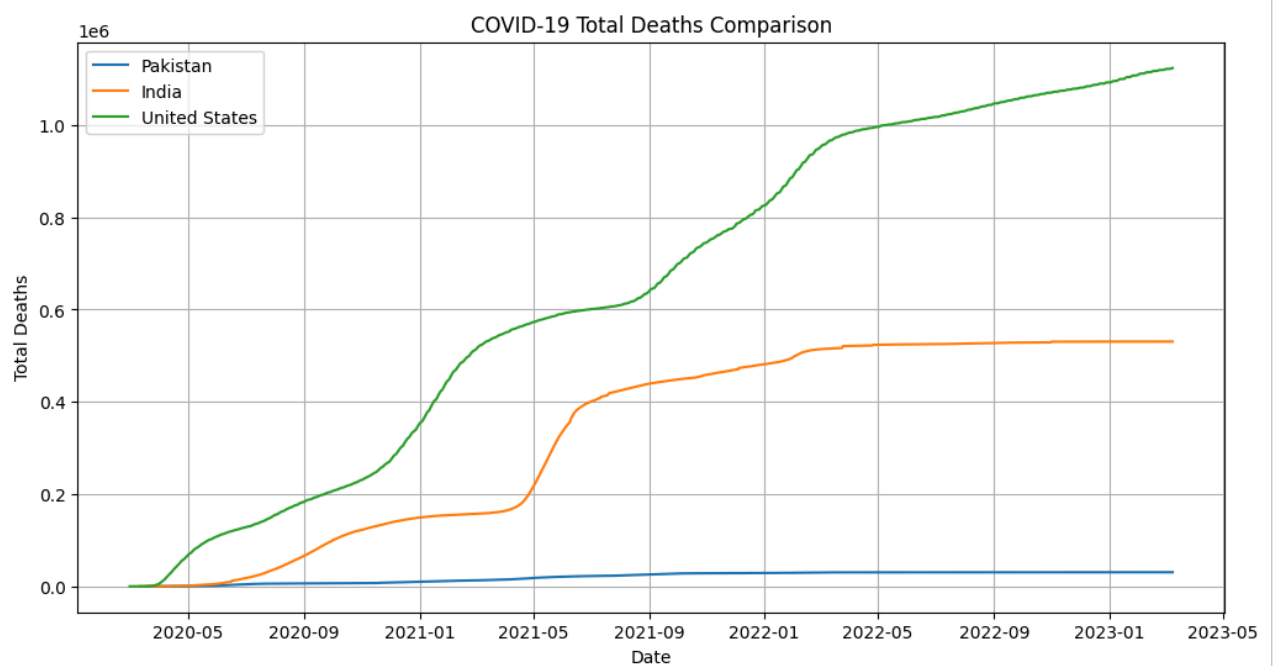
The graph compares cumulative COVID-19 cases across Pakistan, India, and the United States. The United States recorded the highest number of total cases, while Pakistan reported significantly fewer cases throughout the pandemic period.

```
plt.figure(figsize=(12,6))

for country in countries:
    data = selected[selected['location'] == country]
    plt.plot(data['date'], data['total_deaths'], label=country)

plt.title('COVID-19 Total Deaths Comparison')
plt.xlabel('Date')
plt.ylabel('Total Deaths')
plt.legend()
plt.grid(True)

plt.show()
```



✓ Total Deaths Comparison

The death trends generally followed the growth of COVID-19 cases. Countries with larger outbreaks experienced higher cumulative death counts. The United States showed the highest mortality among the selected countries.

```
latest = df.sort_values('date').groupby('location').tail(1)

top10 = latest.sort_values(
    by='total_cases',
    ascending=False
).head(10)

top10[['location', 'total_cases']]
```

	location	total_cases	
245109	World	676467356.0	
99651	High income	431911049.0	
75402	Europe	248452416.0	
13233	Asia	210767366.0	
76544	European Union	184184098.0	
236730	Upper middle income	145057512.0	
167302	North America	122717115.0	
235587	United States	103802702.0	
134531	Lower middle income	97175431.0	
208245	South America	67963678.0	

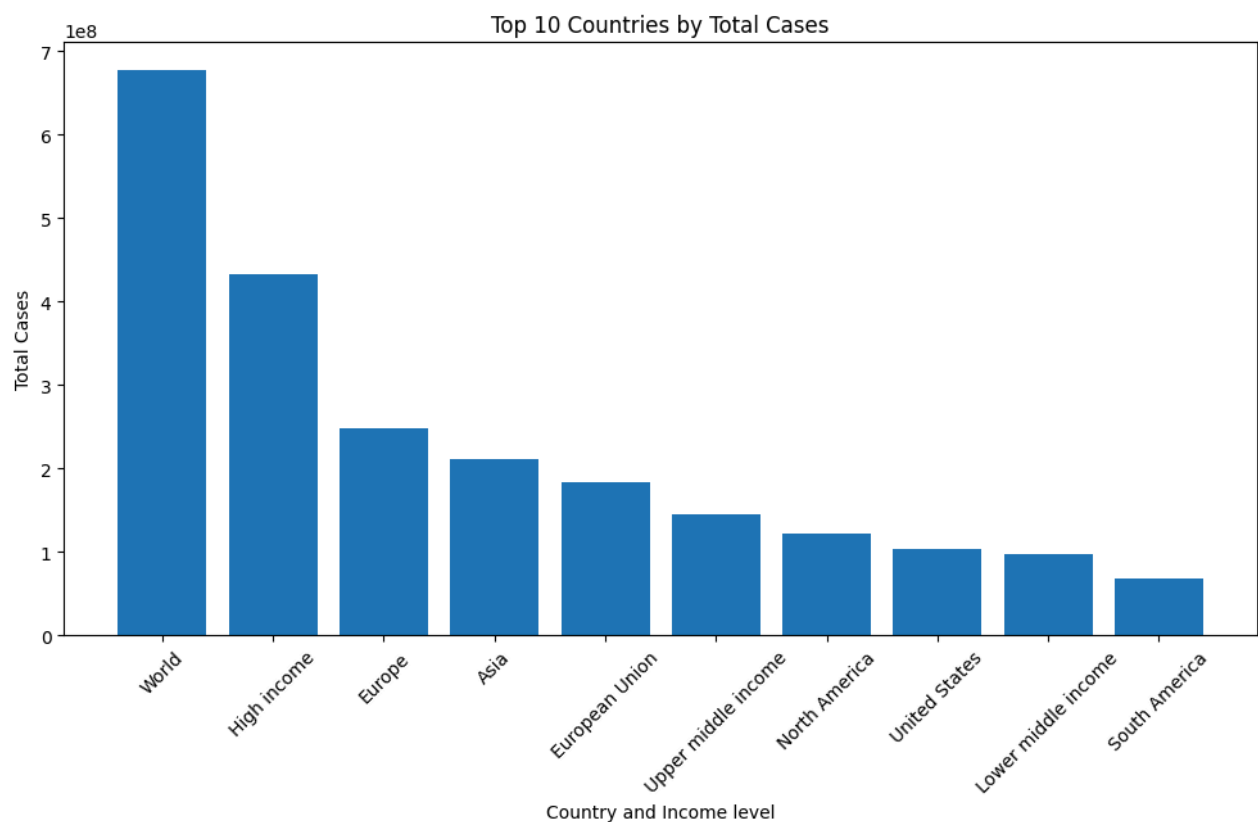
```
plt.figure(figsize=(12,6))

plt.bar(top10['location'], top10['total_cases'])

plt.title('Top 10 Countries by Total Cases')
plt.xlabel('Country and Income level')
plt.ylabel('Total Cases')

plt.xticks(rotation=45)

plt.show()
```



✓ Top Affected Countries

The chart highlights the countries with the largest cumulative COVID-19 case counts in countries and income levels across the world. These countries experienced substantial outbreaks and contributed significantly to global case totals.

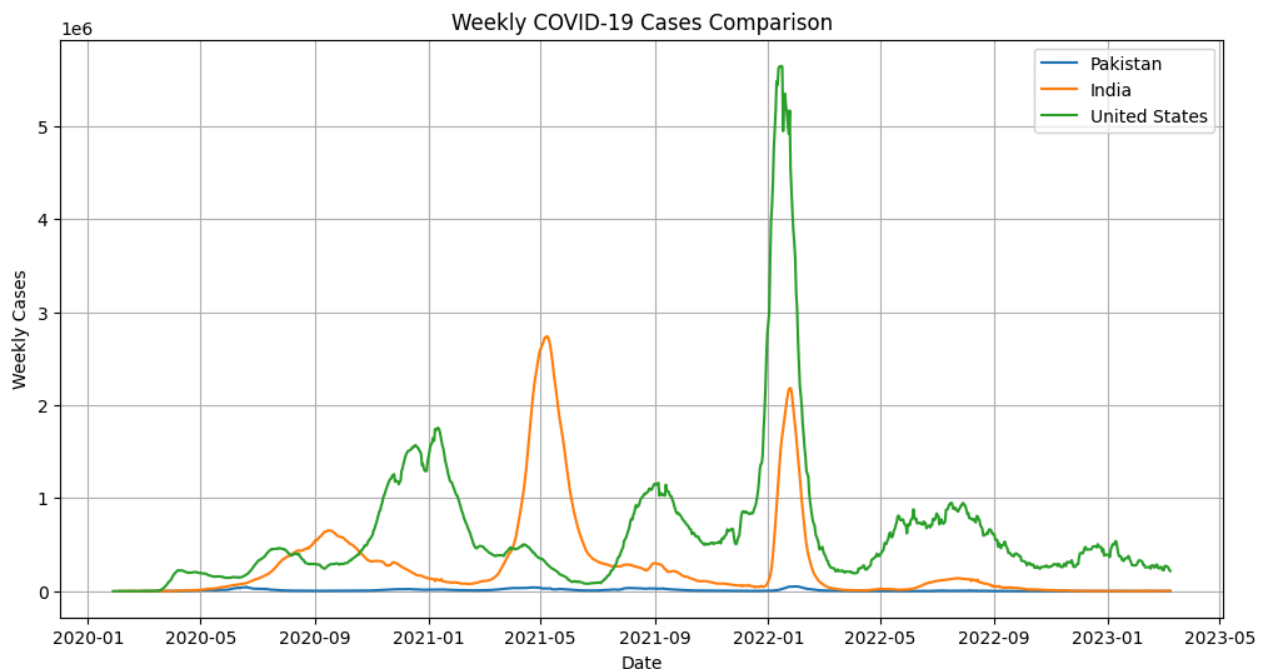
```
plt.figure(figsize=(12,6))

for country in countries:
    data = selected[selected['location'] == country]

    plt.plot(
        data['date'],
        data['weekly_cases'],
        label=country
    )

plt.title('Weekly COVID-19 Cases Comparison')
plt.xlabel('Date')
plt.ylabel('Weekly Cases')
plt.legend()
plt.grid(True)

plt.show()
```



✓ Weekly Cases Trend Analysis

The weekly cases graph highlights the progression of COVID-19 infections over time. Major waves of infection can be identified through sharp increases in weekly reported cases. The chart helps visualize periods of rapid spread and recovery.

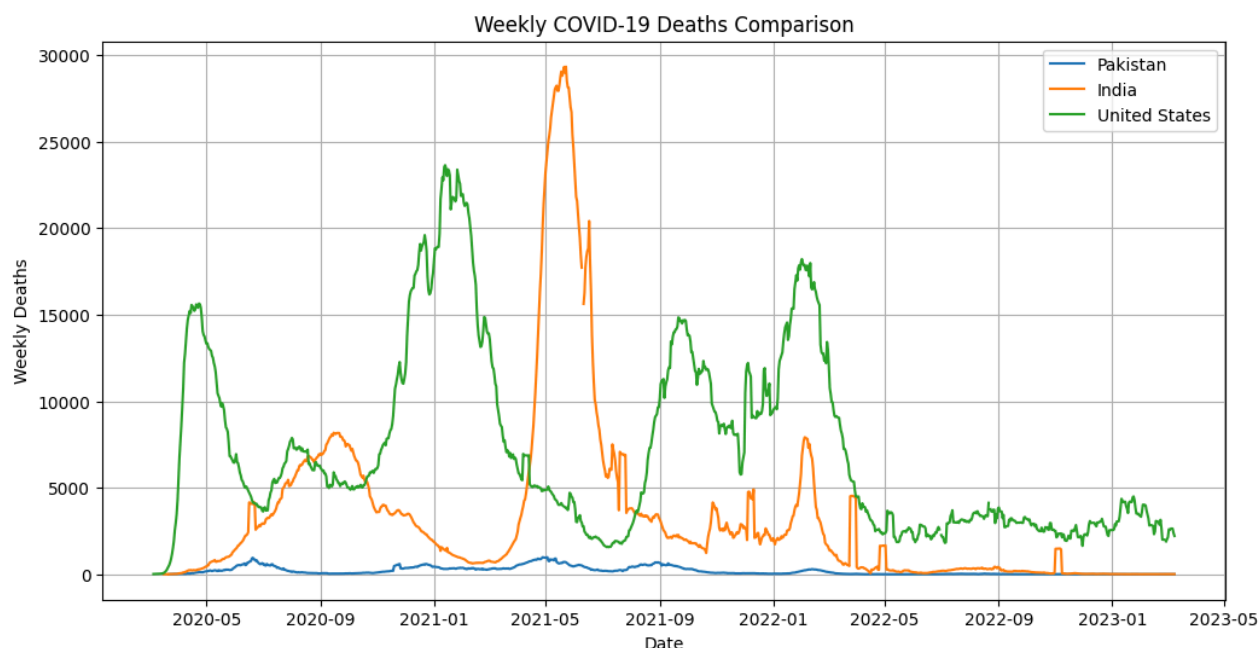
```
plt.figure(figsize=(12,6))

for country in countries:
    data = selected[selected['location'] == country]

    plt.plot(
        data['date'],
        data['weekly_deaths'],
        label=country
    )

plt.title('Weekly COVID-19 Deaths Comparison')
plt.xlabel('Date')
plt.ylabel('Weekly Deaths')
plt.legend()
plt.grid(True)
```

```
plt.show()
```



Weekly Deaths Trend Analysis

The weekly deaths trend follows major COVID-19 outbreaks. Countries experiencing larger infection waves generally reported higher weekly death counts. The chart provides insight into the severity of the pandemic over time. Because of the low population of Pakistan compared to the other two countries.

```
for country in countries:
    data = selected[selected['location'] == country]

    peak = data.loc[data['new_cases'].idxmax()]

    print("Country:", country)
    print("Peak Date:", peak['date'])
    print("Highest Daily Cases:", peak['new_cases'])
    print("-----")
```

```
Country: Pakistan
Peak Date: 2020-06-14 00:00:00
Highest Daily Cases: 12073.0
-----
Country: India
Peak Date: 2021-05-06 00:00:00
Highest Daily Cases: 414188.0
-----
Country: United States
Peak Date: 2022-01-10 00:00:00
Highest Daily Cases: 1354505.0
-----
```

Peak Analysis Explanation

The peak analysis identifies the date on which each selected country reported its highest number of daily COVID-19 cases. These peaks represent major outbreak waves and help highlight periods when transmission was most intense. Comparing peak dates across countries provides insight into how the pandemic progressed differently in each region.

Key Findings

- The United States recorded the highest cumulative COVID-19 cases among the selected countries.
- India experienced several major waves of infections during the pandemic.
- Pakistan reported comparatively lower total case counts.
- Weekly case trends revealed distinct peaks corresponding to outbreak periods.

- Death trends generally followed the rise and fall of confirmed cases.

Conclusion

This project analyzed global COVID-19 trends using publicly available data. Comparisons among Pakistan, India, and the United States revealed significant differences in case and death counts. Time-series visualizations helped identify major waves, peaks, and anomalies throughout the pandemic. The findings demonstrate how data analytics can be used to monitor public health events and support informed decision-making.